Stochastic Gradient Descent

In practice, the evaluation of the gradient

$$D_\theta \hat{R}(\theta_t) = D_\theta \left(\frac{1}{n} \sum_{j=1}^{n} \ell(f_{\theta_t}(x_j), y_j) \right)$$

$$= \frac{1}{n} \sum_{j=1}^{n} D_1 \ell(f_{\theta_t}(x_j), y_j) \cdot D_\theta f_{\theta_t}(x_j)$$

is quite expensive for large n.

Often, people evaluate this gradient on a minibatch $B_t \subseteq \{1, \dots n\}$ of size $b_t \ll n$ (a sample of $\{1, \dots n\}$ of size b_t)

$$\widetilde{D_\theta R_t}(\theta_t) = \frac{1}{b_t} \sum_{j \in B_t} D_\theta \ell(f_{\theta_t}(x_j), y_j) \qquad (SG)$$

Note that this is an unbiased estimator of $D_\theta \hat{R}(\theta_t)$:

$$\mathbb{E}\left(\widetilde{DR}(\theta_t) \right) = \mathbb{E}\left(\frac{1}{b_t} \sum_{B_t} D_\theta \ell(f_{\theta_t}(x_j), y_j) \right)$$

$$\overset{\text{sample w. replacem.}}{=} \frac{1}{b_t} b_t \, \mathbb{E}\left(D_\theta \ell(f_{\theta_t}(x_j), y_j) \right)$$

$$= \frac{1}{n} \sum_{j=1}^{n} D_\theta \ell(f_{\theta_t}(x_j), y_j).$$

with covariance matrix (for $b_t = 1$)

$$\frac{1}{n}\sum_{j=1}^{n}\left(D_\theta \ell(f_{\theta_t}(x_j),y_j) - D\hat{R}(\theta_t)\right)\otimes\left(D_\theta \ell(f_{\theta_t}(x_j),y_j) - D\hat{R}(\theta_t)\right)$$

Then, the Stochastic Gradient descent update reads:

$$\theta_{t+1} = \theta_t + \gamma_t \widetilde{D_\theta R}_t(\theta_t) \qquad \text{(SGD)}$$

$$= \theta_t + \gamma_t D_\theta \hat{R}(\theta_t) + \gamma_t\left(\widetilde{D_\theta R}_t(\theta_t) - D_\theta \hat{R}(\theta_t)\right)$$

Analogously to what was done in the GD setting, we want to approximate the above with the solution of a differential equation. In this case, given the stochastic nature of the update, the natural candidate for a limiting equation is a stochastic differential equation (Langevin dynamics) of the form.

$$d\bar{\theta}_t = b(\bar{\theta}_t)\,dt + \sigma(\bar{\theta}_t)\,dB_t \qquad \text{(SGLD)}$$

where $b(\bar{\theta}_t) \in \mathbb{R}^d$ is the drift coefficient

$\sigma(\bar{\theta}_t) \in \mathbb{R}^{d\times p}$ is the diffusion coefficient

B_t is a p-dimensional brownian motion.

A natural guess for $b(\bar{\theta}_t)$, $\sigma(\bar{\theta}_t)$ would be to match the moments of the SGD update with the ones of SGLD:

- $b(\vartheta) = D_\vartheta \hat{R}(\vartheta)$ assume that $\gamma_t \equiv \gamma \ \forall t$

- $\sigma(\vartheta)\sigma(\vartheta)^T = \Sigma_1(\vartheta) = \gamma \mathbb{E}\left(\left(\widetilde{DR}(\vartheta) - D\hat{R}(\vartheta)\right) \otimes \left(\widetilde{DR}(\vartheta) - D\hat{R}\right)\right)$

Let $R_B(f) = \frac{1}{b} \sum_{j=1}^{b} \ell(f(x_j), y_j)$ and G_α be the class of function with α weak derivatives, all with at most poly growth.

<u>Thm</u>: Let $T > 0$, $N = \lfloor T/\gamma \rfloor$. and let ϑ_k be a realization of (SGD). Further let

- $\bar{R}(\vartheta) = \mathbb{E}_B R_B(f_\vartheta)$ satisfy $\bar{R} \in G_4$
- $D_\vartheta R_B$ is L_B-Lipschitz in ϑ with $\mathbb{E}(L_B^m) < \infty \ \forall m \geq 1$

Then, for every $g \in G_3$

$$\max_{k = 0, \dots N} \mathbb{E} g(\bar{\vartheta}_{k\gamma}) - \mathbb{E} g(\vartheta_k) \leq C \gamma$$

Proof: By regularity of $\hat{R}(\vartheta)$, $\sigma(\vartheta)$ the solution $\bar{\vartheta}_t$ exists and is unique.

Now, letting $u(\vartheta, s) = \mathbb{E}(g(\bar{\vartheta}_{k\gamma}) \mid \bar{\vartheta}_{s\gamma} = \vartheta)$

$$|\mathbb{E} g(\bar{\vartheta}_{k\gamma}) - \mathbb{E} g(\vartheta_k)| = \left|\sum_{\ell=1}^{k} \mathbb{E}\mathbb{E}(g(\bar{\vartheta}_{k\gamma}) \mid \bar{\vartheta}_{\ell\gamma} = \vartheta_{\ell-1}) - \mathbb{E}\mathbb{E}(g(\bar{\vartheta}_{k\gamma}) \mid \bar{\vartheta}_{\ell\gamma} = \vartheta_\ell)\right|$$

$$\overset{J}{\leq} \sum_{\ell=1}^{k} \mathbb{E} \left| \mathbb{E}(u(\bar{\vartheta}_{\ell\gamma}, \ell\gamma) \mid \bar{\vartheta}_{(\ell-1)\gamma} = \vartheta_{\ell-1}) - \mathbb{E}(u(\vartheta_\ell, \ell\gamma) \mid \vartheta_{\ell-1} = \vartheta_{\ell-1})\right|$$

$$= \sum_{\ell=1}^{k} \mathbb{E}\left|\mathbb{E}\left(u(\bar\vartheta_{\ell\gamma}, \ell\gamma) - u(\vartheta_\ell, \ell\gamma) \mid \vartheta_{\ell-1} = \bar\vartheta_{(\ell-1)\gamma}\right)\right|$$

$$\lesssim \sum_{\ell=1}^{k} \mathbb{E}\left|\mathbb{E}\left(D_\vartheta u(\vartheta_{\ell-1})(\bar\vartheta_{\ell\gamma} - \vartheta_\ell) \mid \vartheta_{\ell-1} = \bar\vartheta_{(\ell-1)\gamma}\right)\right| + O(\gamma^2)$$

$$\le \sum_\ell \mathbb{E}\left|K(\vartheta_{\ell-1})\, \mathbb{E}\left(\bar\vartheta_{\ell\gamma} - \vartheta_\ell \mid \vartheta_{\ell-1} = \bar\vartheta_{\ell-1}\right)\right| + O(\gamma^2)$$

$$\le \sum_\ell \mathbb{E}\Big(K(\vartheta_{\ell-1})\, \mathbb{E}\Big(\cancel{\vartheta_{\ell-1}} + \int_{(\ell-1)\gamma}^{\ell\gamma} D_\vartheta \hat R(\bar\vartheta_s)\, ds + \cancel{\sqrt{\gamma}\int_{(\ell-1)\gamma}^{\ell\gamma} \sigma(\bar\vartheta_s)\, dB_s} \quad \mathbb{E}$$

$$- \left(\cancel{\vartheta_{\ell-1}} + \gamma D_\vartheta \hat R(\vartheta_{\ell-1})\right)\Big)\Big) + O(\gamma^2) \mid \bar\vartheta_{(\ell-1)\gamma} = \vartheta_{\ell-1}\Big)$$

itô

$$\le \sum_\ell \mathbb{E}\Big(K(\vartheta_{\ell-1})\, \mathbb{E}\Big(\gamma D_\vartheta \hat R(\bar\vartheta_{(\ell-1)\gamma}) + \int_{(\ell-1)\gamma}^{\ell\gamma}\int_{(\ell-1)\gamma}^{s} D_\vartheta \hat R(\bar\vartheta_r) D^2_\vartheta \hat R(\bar\vartheta_r)\, dr\, ds$$

$$+ \underbrace{\iint dW_r\, ds}_{0} - \gamma D_\vartheta \hat R(\vartheta_{\ell-1}) \mid \bar\vartheta_{(\ell-1)\gamma} = \vartheta_{\ell-1}\Big) + O(\gamma^2)$$

$$\le O(\gamma^2) \sum_\ell \mathbb{E}\left(K(\vartheta_{\ell-1})\right)$$

Having proven finite time! convergence of the SGD dynamics to SGLD we want to give sufficient conditions for convergence to minimizers of SGLD:

The distribution at time t $P(\bar\vartheta_t \in d\vartheta) = \rho_t(\vartheta)\, d\vartheta$ obeys the Fokker-Planck equation

$$\partial_t \rho_t(\vartheta) = -\operatorname{div}\left(\rho_t(\vartheta) D_\vartheta \hat R(\vartheta)\right) + \frac{1}{2}\sqrt{\eta}\, D^2_\vartheta : \left(\Sigma(\vartheta)\rho_t(\vartheta)\right)$$

where $D^2_\vartheta : A(\vartheta) = \sum_{j=1}^{d}\sum_{i=1}^{d} \frac{\partial}{\partial\vartheta_i}\frac{\partial}{\partial\vartheta_j} A_{ij}(\vartheta)$

The convergence to equilibrium depends on the degeneracy of $\Sigma(\vartheta)$: if $\Sigma(\vartheta)$ is nondegenerate ($n \gg d$) then.

<u>Thm</u>: Let $\Sigma(\vartheta) \succeq \lambda \mathbb{1}$ for all $\vartheta \in \Theta$ and let $\hat{R}(\vartheta)$ be λ-scfa $\lambda > 0$ and C^1 then there exists a density ρ_* such that

$$D_{KL}(\rho_t | \rho_*) \leq e^{-\lambda t} D_{KL}(\rho_0 | \rho_*)$$

where $D_{KL}(\rho | \rho') = \int_\Theta \log \frac{\rho'(\vartheta)}{\rho(\vartheta)} \rho(\vartheta) \, d\vartheta$ is the Kullback Leibler divergence or relative entropy.

However, in reality we cannot evaluate $\mathbb{E}(g(\bar{\vartheta}_t))$...

<u>Proposition</u>: assume that R is convex, L-Lipschitz and that $\|\tilde{D}R - D\hat{R}\|$ is uniformly bounded a.s. on Θ. Then choosing $\eta_t \approx \mathcal{O}(\frac{1}{\sqrt{t}})$ we have (neglecting log terms)

$$\mathbb{E}(\hat{R}(\tilde{\vartheta}_t) - \hat{R}(\vartheta_*)) \leq K \frac{\|\vartheta_0 - \vartheta_*\|}{\sqrt{t}}$$

where $\hat{\vartheta}_t := \frac{\sum_{s=1}^{t} \gamma_s \vartheta_s}{\sum_{s=1}^{t} \gamma_s}$

Note: if $d \gg n$ we will be able to interpolate the data: $\ell(f_{\vartheta^*}(x_i), y_i) = 0 \ \forall i$

$\Rightarrow$ when $\hat{R}(\vartheta_*) = 0$ we also have $\Sigma(\vartheta^*) = 0$ $\Rightarrow$ there are fixed points for SGD!

<u>Summary</u>: GD $\xrightarrow{\gamma \to 0}$ GF

SGD $\xrightarrow{\gamma \to 0}$ SGLD

	convex	λsc
GD	$1/t$	$e^{-\lambda t}$
SGD	$1/\sqrt{t}$ $\left(\gamma_t = \frac{1}{\sqrt{t}}\right)$	$1/t$ $\left(\gamma = \frac{1}{t}\right)$

<u>However</u>: underlying assumption: convexity in θ

This assumption is <u>not</u> respected by NN:

Assume $\theta_* = (w_1^*, w_2^*, a_1^*, a_2^*)$ is minimizer of $\hat{R}$

$$f_{\theta_*}(x) = a_1^* \sigma(w_1^* x) + a_2^* \sigma(w_2^* x)$$

$$= a_2^* \sigma(w_2^* x) + a_1^* \sigma(w_1^* x) = f_{\theta_*'}(x)$$

$$\theta_* = (w_1^*, w_2^*, a_1^*, a_2^*) \neq (w_2^*, w_1^*, a_2^*, a_1^*) = \theta_*'$$

no reason why on $\alpha \theta_* + (1-\alpha)\theta_*'$ $\hat{R}$ should be minimized $\rightarrow$ <u>nonconvexity</u> !!